

SOVEREIGN INDUSTRIAL COMPLIANCE REPORT

EQUIPMENT INSPECTION DOSSIER: PRESSURE VESSEL PS-26117

Classification: SYNTHETIC DEMONSTRATION RECORD | Inspection Date: 2026-09-12 | Inspector ID: INS-8419

Asset Tag	PS-26117	Asset Type	Vertical Hydrotreater Separator
Location	Unit 4 Hydrotreating Complex	Design Code	ASME Section VIII Div 1
Design Pressure (MAWP)	100.0 PSI (689 kPa)	Design Temperature	350°F (176.7°C)
Hydrostatic Test Pressure	150.0 PSI (1.5x MAWP)	Test Date	2026-09-12 11:30 UTC
Nominal Wall Thickness	15.0 mm	Minimum Measured Thickness	14.2 mm (Shell Section 2)
Ultrasonic NDT Status	Grade 1 (Full Penetration)	Weld Joints Examined	Circumferential & Long. Seams
Relief Valve PRV-261	150.0 PSI (Calibrated/Passed)	Popping Tolerance	+/- 1.5% compliant
Physical QA Stamp	PENDING VERIFICATION	Inspector Recommendation	CONDITIONAL RECERTIFICATION

1.0 Ultrasonic Shell Thickness Profile

Precision ultrasonic examination was performed across 16 calibrated grid coordinates. The minimum recorded shell wall thickness was 14.2 mm on lower course shell section 2. All measured coordinates exceed the 12.0 mm retirement threshold established under SOP-704.

2.0 Weld Integrity & Hydrostatic Verification

All head-to-shell circumferential welds and shell longitudinal seam welds were inspected using ultrasonic phased-array testing. Zero crack-like indications, lack of fusion, or porosity clusters were observed. Hydrostatic proof pressure was held at 150 PSI for 60 minutes with zero observable pressure drop.

3.0 Finding & Compliance Action Item

CRITICAL NOTICE: While physical mechanical tests comply with ASME standards, the secondary inspection QA stamp has not yet been physically verified on the vessel nameplate. Approval must remain **CONDITIONAL** until the on-site physical stamp is confirmed.